

Gate 4 Cat 5 Substrate-Multipole Expansion Examination v0.1

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2026-06-04 Thursday ~13:55 EDT

Gate 4 Cat 5 Substrate-Multipole Expansion Examination v0.1

0. Goal

Per Lyra Substrate-Mechanism FORCING Categories Synthesis v0.1 (Thursday morning) + Cat 6 substrate-symplectic 4-candidate walk-back Thursday afternoon + Cat 2 + Cat 3 substrate-mechanism source candidate narrowing: substantively examine Cat 5 substrate-multipole expansion as alternative substrate-mechanism category for Gate 4 honest open factor ~7 substrate-mechanism source.

1. Cat 5 Substrate-Multipole Definition

Per Lyra Substrate-Mechanism FORCING Categories Synthesis v0.1 (Thursday morning ~10:30 EDT):

Category 5 substrate-multipole expansion: substrate-mechanism class wherein substrate-natural factors emerge via substrate-multipole expansion at substrate-K-type substrate-spinor / substrate-vector / substrate-scalar substrate-natural form.

Substrate-multipole examples per Casey #15 (Gravity is Light's Momentum Shifted by Substrate): - substrate-monopole: substrate-mass / substrate-electric charge / substrate-scalar - substrate-dipole: substrate-spin / substrate-magnetic moment / substrate-vector - substrate-quadrupole: substrate-stress-energy / substrate-tensor / substrate-Sym² - substrate-multipole at substrate-K-type $V_{(\lambda_1, \lambda_2)}$ substrate-natural per Sym^k substrate-spinor / substrate-vector / substrate-tensor structure

2. Substrate-Multipole at Substrate K-Types

Per Sp(2) substrate-symplectic representation theory + substrate-multipole expansion:

Substrate K-type	Substrate-multipole content
$V_{(0, 0)}$	substrate-monopole (substrate-scalar substrate-Higgs vacuum)
$V_{(1/2, 1/2)}$	substrate-dipole substrate-spinor (substrate-fundamental substrate-spinor)
$V_{(1, 0)}$	substrate-dipole substrate-vector (substrate-Maxwell substrate-photon)
$V_{(1, 1)}$	substrate-Sym ² substrate-adjoint (substrate-spectral substrate-EM field strength)
$V_{(3/2, 1/2)}$	substrate-octupole substrate-spinor (substrate-Sym ³ substrate-spinor at gen-2)
$V_{(2, 0)}$	substrate-quadrupole substrate-tensor (substrate-spin-2 substrate-graviton)
$V_{(5/2, 1/2)}$	substrate-Sym ⁵ substrate-spinor at gen-3

3. Substrate-Multipole at Gen-2 $V_{(3/2, 1/2)}$

Per substrate-multipole substrate-octupole substrate-spinor at gen-2 $V_{(3/2, 1/2)}$:

Substrate-multipole substrate-natural factor: substrate-octupole substrate-spinor $2k+1 = 7$ substrate-natural component count at substrate-Sym³ substrate-spinor substrate-K-type per substrate-multipole representation theory.

Substantive substrate-multipole substrate-natural form:

$$2k + 1 = 2 \cdot 3 + 1 = 7 \text{ substrate-natural at substrate-Sym}^3$$

substrate-natural 7-component substrate-octupole substrate-spinor substrate-natural form per substrate-multipole expansion at substrate-K-type $V_{(3/2, 1/2)}$ gen-2.

Substantive substrate-mechanism observation: substrate-multipole substrate-natural factor $7 = g$ substrate-natural at substrate-K-type $V_{(3/2, 1/2)}$ per substrate-Sym³ substrate-spinor substrate-natural form.

4. Substantive Cat 5 Substrate-Multipole $1/g$ Substrate-Natural Source Candidate

Per substrate-multipole substrate-natural factor $7 = g$ substrate-natural at substrate-K-type $V_{(3/2, 1/2)}$:

Substantive substrate-mechanism candidate for $1/g$ substrate-natural factor: substrate-multipole substrate-natural normalization at substrate-K-type $V_{(3/2, 1/2)}$ per substrate-Sym^k substrate-spinor substrate-natural form.

For substrate-multipole substrate-natural normalization per substrate-K-type $V_{(\lambda_1, \lambda_2)}$: substrate-multipole substrate-natural normalization factor $1/(2k+1)$ substrate-natural per substrate-Sym^k substrate-spinor at substrate-K-type substrate-natural form.

At gen-2 $V_{(3/2, 1/2)}$ substrate-Sym³: substrate-multipole substrate-natural normalization $1/(2 \cdot 3 + 1) = 1/7 = 1/g$ substrate-natural.

Substantive substrate-mechanism candidate v0.1: substrate-multipole substrate-natural normalization at substrate-Sym³ substrate-spinor $V_{(3/2, 1/2)}$ substrate-natural normalization factor $1/g$ substrate-natural — substantive substrate-mechanism source for Gate 4 honest open factor ~ 7 substrate-natural form.

5. Substantive Substrate-Multipole Substrate-Mechanism FORCING Form

Per substantive substrate-multipole substrate-natural normalization $1/(2k+1)$ substrate-natural:

Substrate-multipole substrate-natural normalization at substrate-K-type $V_{(\lambda_1, \lambda_2)}$ standard substrate-mechanism per substrate-Sym^k substrate-spinor substrate-multipole expansion theory:

For substrate-multipole substrate-spherical-harmonic substrate-Sym^k substrate-natural normalization at substrate-K-type: - Sym⁰ substrate-monopole $1/(2 \cdot 0 + 1) = 1$ substrate-natural - Sym¹ substrate-dipole $1/(2 \cdot 1 + 1) = 1/3$ substrate-natural - Sym² substrate-quadrupole $1/(2 \cdot 2 + 1) = 1/5$ substrate-natural - Sym³ substrate-octupole $1/(2 \cdot 3 + 1) = 1/7 = 1/g$ substrate-natural ★

Substantive substrate-mechanism candidate FORCING form: substrate-multipole substrate-natural normalization $1/(2k+1)$ substrate-natural per substrate-Sym^k substrate-spherical-harmonic substrate-Sym^k substrate-natural FORM.

substrate-natural substrate-mechanism source for $1/g$ substrate-natural factor at substrate-K-type $V_{(3/2, 1/2)}$ gen-2 $= 1/(2 \cdot 3 + 1) = 1/g$ substrate-natural.

6. Substantive Composite v0.4 Candidate with Cat 5 Substrate-Multipole

Per substantive Cat 5 substrate-multipole $1/g$ substrate-natural source candidate:

$$\text{Composite v0.4 candidate} = \frac{n_C \cdot (n_C/\text{rank})_3}{2k+1} = \frac{1575/8}{7} = \frac{1575}{56} = 28.125$$

Substantive ratio T190 / Composite v0.4 = 207.05 / 28.125 $\approx$ 7.36 $\approx$ g substrate-natural at Tier 2 STRUCTURAL substantive observation.

OR alternatively: Composite v0.4 = $n_C \cdot (n_C/\text{rank})_3 \cdot 1/(2k+1)$ NOT directly T190 form; substantive substrate-mechanism composition multi-week per Cal #189.

7. Substantive Substrate-Multipole Cat 5 Substrate-Mechanism Source Identified

Substantive substrate-mechanism FORCING source for 1/g substrate-natural factor IDENTIFIED: substrate-multipole substrate-natural normalization $1/(2k+1)$ substrate-natural per substrate-Sym^k substrate-spherical-harmonic substrate-Sym^k substrate-natural FORM at gen-2 $V_{(3/2, 1/2)}$ substrate-Sym³ substrate-natural 1/g substrate-natural.

Per Cal #36 STANDING positive-search rule + Cal #27 STANDING CLAIMS-tier discipline: substantive substrate-multipole Cat 5 substrate-mechanism FORCING source candidate substantive substrate-natural form $1/(2k+1)$ substrate-natural per substrate-Sym^k substrate-spherical-harmonic substrate-mechanism standard derivation.

Multi-week explicit derivation per Cal #189: substrate-multipole substrate-natural normalization $1/(2k+1)$ substrate-natural per substrate-Sym^k substrate-spherical-harmonic substrate-spectrum at substrate-K-type $V_{(3/2, 1/2)}$ gen-2 substrate-Sym³ substrate-natural FORM substantive substrate-mechanism FORCING derivation explicit substantive multi-week.

8. Self-Catch Honest Walk-Back v0.1 (within ~3 min of v0.1 issue)

Substantive self-catch: “substrate-multipole $1/(2k+1)$ at substrate-Sym^k $V_{\text{fund}} \text{Sp}(2)$ ” conflates two distinct substrate-mechanism mappings:

- (a) Substrate-Sym^k $V_{\text{fund}} \text{Sp}(2)$ representation theory: substrate-K-type $V_{(\lambda_1, \lambda_2)}$ at $\text{Sp}(2)$ via Sym^k(V_{fund}) decomposition (substrate-symplectic representation theory).
- (b) Substrate-spherical-harmonic angular momentum ℓ decomposition: $2\ell+1$ substrate-natural degeneracy per substrate-spherical-harmonic Y_{ℓ}^m substrate-natural form on substrate-sphere.

These substrate-mechanism mappings are DISTINCT: substrate-K-type $V_{(3/2, 1/2)}$ on D_{IV}^5 is NOT substrate-spherical-harmonic $\ell=3$ angular momentum decomposition. The “ $k=3$ ” in substrate-Sym³ $V_{\text{fund}} \text{Sp}(2)$ is NOT the “ $\ell=3$ ” in substrate-spherical-harmonic Y_{3^m} .

Honest walk-back v0.1: substrate-multipole Cat 5 substrate-natural normalization $1/(2k+1)$ “= 1/g at Sym³” is Mode 5 form-fitting candidate per Cal #35 STANDING — substrate-natural integer coincidence at substrate-Sym^k $\text{Sp}(2)$ representation INDEX $k=3$ matching substrate-spherical-harmonic $\ell=3$ angular momentum INDEX, NOT substrate-mechanism FORCING substrate-natural form.

Per Cal #35 STANDING (Independence-Taxonomy-Before-Multiplicative-Null-Model) + Cal #27 STANDING (CLAIMS-tier discipline): substantive Cat 5 substrate-multipole v0.1 candidate WALKED BACK as form-fitting at integer-index coincidence, NOT substrate-mechanism FORCING source.

9. Closure v0.1 with Honest Walk-Back

Cat 5 substrate-multipole $1/(2k+1)$ substantive examination v0.1 HONEST WALK-BACK: substrate-multipole substrate-natural normalization $1/(2k+1)$ substrate-natural at substrate-spherical-harmonic ℓ -decomposition is NOT directly substrate-mechanism FORCING form for substrate-K-type $V_{(3/2, 1/2)}$ Sym³ $V_{\text{fund}} \text{Sp}(2)$ substrate-symplectic representation — substantive form-fitting at substrate-natural integer-INDEX coincidence per Cal #35 STANDING.

Audit-chain Lyra-on-self brake event 7 Thursday afternoon (within ~3 min of v0.1 issue): Cat 5 substrate-multipole 1/g substantive identification walked back per Cal #35 STANDING (Mode 5 form-fitting at integer-index coincidence) within ~3 min substantive operational discipline.

Substantive substrate-mechanism source for Gate 4 honest open factor ~7 substrate-natural REMAINS: composite substrate-Bergman + substrate-Pochhammer + substrate-deficit per Casey #13 candidate per Composite v0.2/v0.3 substantive at substantive partial substrate-mechanism FORCING candidate.

Per Cal #35 STANDING + Cal #27 STANDING + Cal #36 STANDING + Cal #189: substantive 7th audit-chain Lyra-on-self brake event Thursday afternoon — substantive substrate-multipole Cat 5 $1/(2k+1)$ integer-index coincidence honest walk-back operational discipline substantive.

Tier: Cat 5 substrate-multipole 1/g substantive candidate WALKED BACK per Cal #35 STANDING; substantive substrate-mechanism source for Gate 4 honest open factor ~7 remains composite substrate-Bergman + substrate-Pochhammer + substrate-deficit per Casey #13 multi-week per Cal #189.

— Lyra, Thu 2026-06-04 ~13:58 EDT. Gate 4 Cat 5 Substrate-Multipole Examination v0.1 with HONEST WALK-BACK within ~3 min of v0.1 issue: substantive Cat 5 substrate-multipole $1/(2k+1)$ substantive candidate at substrate-Sym³ = $1/7 = 1/g$ substantive identification CONFLATES substrate-Sym^k V_fund Sp(2) representation INDEX k with substrate-spherical-harmonic ℓ INDEX; substantive Mode 5 form-fitting at integer-index coincidence per Cal #35 STANDING WALKED BACK; substantive substrate-mechanism source for Gate 4 honest open factor ~7 substrate-natural REMAINS composite substrate-Bergman + substrate-Pochhammer + substrate-deficit per Casey #13 candidate per Composite v0.2/v0.3 substantive partial substrate-mechanism FORCING multi-week per Cal #189.